

OPERATING SYSTEM COMPARISON

ITEP 414 - System Administration and Maintenance | Week 3

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Course/Year: BSIT 4C

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Date: August 15, 2026

Comparative Table

Aspect	Windows Server	Ubuntu Server	Rocky Linux
Licensing	Commercial. Sold per core with different editions (Standard, Datacenter). CALs may be required for clients.	Free and open source, released under the GPL. Anyone can download, use, and modify it at no cost.	Free and open source. A community rebuild of Red Hat Enterprise Linux, so it has no subscription cost.
User Interface	Full graphical desktop and Server Manager GUI, plus PowerShell. Very beginner-friendly to navigate.	Command line by default. A minimal desktop can be added, but servers are managed through the terminal.	Command line by default, same as most Linux servers. Can also add a GUI if really needed.
Package Management	MSI/MSIX installers and Windows Server Update Services (WSUS). No true package repository like Linux.	APT with the .deb format. Thousands of packages in the Ubuntu repositories, installed with apt install.	DNF with the .rpm format. Draws packages from the RHEL-compatible Rocky repositories.
Security	Secure Boot, BitLocker, Defender, and regular patch cycles, but a common target for malware because of market share.	Open source code that gets audited by many people. Uses AppArmor, UFW, and has a fast security update stream.	Inherits RHEL hardening with SELinux enabled by default, one of the strongest mandatory access control systems.
Performance	Solid performance, though heavier because of the GUI and background services.	Very light. A minimal install uses well under 1 GB of RAM, leaving more room for applications.	Also very light. A minimal install stays small and fast, similar to Ubuntu but RHEL-based.
Typical Enterprise Use	Active Directory, Exchange, SQL Server, and shops that are already all-Windows.	Web servers, databases, cloud workloads, and the default choice for most startups and developers.	Same workloads as RHEL: production web and database servers in companies that want RHEL compatibility without the bill.
Advantages	Great support, Active Directory, and a huge ecosystem of commercial software and certified admins.	Free, huge community, tons of documentation, easy automation, and LTS releases with 5 years of support.	RHEL compatibility, SELinux, very stable, and free. Preferred where stability matters more than bleeding edge features.

Disadvantages	Licensing costs add up, GUI eats resources, and patch management needs discipline.	Command line learning curve, and vendor support is community-driven unless you pay Canonical for it.	Smaller community than Ubuntu, packages can be older, and fewer tutorials are written for it.
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Discussion

For this project I installed Ubuntu Server, but in the Week 3 module I also had to study Windows Server and Rocky Linux side by side. The three systems approach the same job very differently, and the comparison shows why a company like our startup scenario would pick one over the others.

Money is usually the first thing a startup looks at, and that is where the Linux side wins. Ubuntu Server and Rocky Linux are free, while Windows Server charges per core and often adds Client Access Licenses on top. A small company with one or two servers can save thousands of pesos a year by going with Linux. On the other hand, if the company already runs Active Directory, Exchange, or SQL Server, Windows is almost a must because of those services, and the licensing becomes easier to justify as a business cost.

Day to day, the difference is in how the administrator works. Windows Server gives you a graphical Server Manager and PowerShell, which feels familiar to anyone coming from a desktop Windows background. Ubuntu and Rocky are terminals first. Commands like `apt install`, `systemctl status`, and `vim` become your daily tools. It is a steeper start, but once learned, a text-based server is faster to manage, easier to script, and much lighter on resources, which is why the Ubuntu Server I installed here uses only a fraction of the memory a Windows install would use.

Security and stability also separate them. Windows is hardened but remains the biggest attack target simply because of market share. Ubuntu uses AppArmor and pushes security updates quickly. Rocky Linux comes from the RHEL line, so it ships with SELinux on by default, which is the strongest access control of the three. For a company that runs public web services, Rocky is often chosen for that extra enforcement, while Ubuntu is chosen for its enormous community, up-to-date packages, and the fact that almost every tutorial on the internet is written for it.

In the end there is no single best system, only the right fit. For our startup scenario, Ubuntu Server made sense because of the zero licensing cost, the lighter footprint, and the massive community support. A company with strict compliance needs or heavy Microsoft integration would reasonably go with Rocky Linux or Windows Server instead. Knowing all three is what makes a system administrator useful, because the real skill is matching the operating system to the business need.

References

1. Microsoft. Windows Server 2022 documentation. <https://learn.microsoft.com/windows-server>
2. Canonical. Ubuntu Server documentation. <https://ubuntu.com/server/docs>
3. Rocky Linux. Documentation. <https://docs.rockylinux.org>
4. CompTIA. Server+ (SK0-005) Exam Objectives. <https://www.comptia.org/certifications/server>